

CertTalk: A Clear, Proof-First Protocol for Two Agents With Private Maps

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Abstract

Two agents each see only part of a grid’s obstacles. They must decide if a route exists from start s to goal t in the union map U —and hand over a proof anyone can replay later. CertTalk does not ship maps; it ships a certificate. If a path exists, the certificate is the path. If no path exists, the certificate is a small set of blocked cells that separates s from t . This writeup presents the idea in plain language, shows two worked examples, and prints full, annotated transcripts for the baselines so a reader can verify what each system actually sends.

1 What We Decide (and What We Don’t)

Problem. A knows obstacles A . B knows obstacles B . Neither sees the union $U = A \cup B$. The question is: does a 4-neighbour path exist from s to t in U ?

Contract. The dialogue ends with a single certificate that a third-party checker (the “oracle”) can replay using only the certificate, s , t , grid size, and U . The oracle never reads the negotiation transcript and never needs the agents’ planner code.

Out of scope. No motion control, dynamics, or multi-agent coordination beyond the yes/no reachability proof. Certificates are short (target ≤ 256 bytes) and archivable.

2 How the Proof Works

Path certificate (checker’s steps).

1. Decode Base64 runs (RLE4 bytes).
2. Verify `digest16` (CRC16 of raw bytes).
3. Expand runs into coordinates from s to t ; check 4-connectivity and bounds.
4. Confirm no coordinate lies in a blocked cell of U .
5. Accept. The bytes *are* the proof.

Cut certificate (checker’s steps).

1. Decode Base64 cell list (DELTA16) and witness bits.
2. Verify `digest16` (CRC16 of raw bytes).
3. Check each listed cell is blocked in U .
4. Remove the cells from the underlying 4-neighbour graph; verify s and t are disconnected.
5. Accept. The bytes *are* the proof.

Witness bits. One bit per cut cell says “who saw this cell” (0=A, 1=B). They guide the responder during negotiation. The oracle ignores witness semantics except to check length.

3 Worked Example: Reachable (seed 247)

When a path exists, A proposes a concrete route. B replays it locally. If every step is free for B, B upgrades it to a certificate. Three messages end the dialogue.

CertTalk — PATH success

```

1 {
2   "t": "PATH_PROPOSE",
3   "s": "v1", "q": 0,
4   "p": { "runs": "QoVBgkKCRA==", "digest16": 22270 },
5   "c": 16269
6 }
7 {
8   "t": "PATH_CERT",
9   "s": "v1", "q": 0,
10  "p": { "runs": "QoVBgkKCRA==", "digest16": 22270, "signed_by": ["B"] },
11  "c": 49459
12 }
13 {
14   "t": "ACK",
15   "s": "v1", "q": 1,
16   "p": { "ack_of": "PATH_CERT", "digest16": 22270 },
17   "c": 3754
18 }
```

What B checks. Decode runs; verify CRC; expand steps; confirm every coordinate is free in B’s map; certify. The certificate itself is the replayable proof.

4 Worked Example: Unreachable (seed 145)

When no path exists, A proposes a short list of blocked cells that separates s from t . B validates per-cell (using witness attribution to allow “I saw it” from A) and then validates disconnection. Three messages suffice on this seed.

CertTalk — CUT success

```

1 {
2   "t": "CUT_PROPOSE",
3   "s": "v1", "q": 0,
4   "p": { "cs": "AAACAAH/Af8=", "k": 3, "digest16": 11880, "w": "AA==" },
5   "c": 51694
6 }
7 {
8   "t": "CUT_CERT",
9   "s": "v1", "q": 0,
10  "p": { "cs": "AAACAAH/Af8=", "k": 3, "digest16": 11880, "w": "AA==", "sb": ["B"] },
11  "c": 43260
12 }
13 {
14  "t": "ACK",
15  "s": "v1", "q": 1,
16  "p": { "ack_of": "CUT_CERT", "digest16": 11880 },
17  "c": 27687
18 }

```

What B checks. Decode cells and witness; verify CRC; accept each cell (blocked locally or credited to A); remove the set and confirm *s* and *t* are disconnected; certify.

5 Baselines: What They Send (and Why It Matters)

Different systems reach the same decision with different bytes. Below we show one clean unreachable seed (145) where all four succeed, and why a reader should care.

Send-All — CUT success (seed 145)

```

1 {
2   "type": "PROBE", "schema": "v1", "seq": 0,
3   "payload": { "what": "CELLS",
4     "indices":
5       [2,3,4,11,13,14,17,19,20,22,27,32,36,37,38,39,46,47,63,65,68,70,71,78,80,83,85,95,98],
6     "chunk": 0, "total": 1, "final": true },
7   "crc16": 40037
8 }
9 {
10  "type": "CUT_CERT", "schema": "v1", "seq": 0,
11  "payload": { "cells": "AgAAP4BAQA=", "encoding": "DELTA16_v1",
12    "digest16": 34702, "k": 3, "witness": "Ag==", "signed_by": ["B"] },
13  "crc16": 12256
14 }
15 {
16  "type": "ACK", "schema": "v1", "seq": 1,
17  "payload": { "ack_of": "CUT_CERT", "digest16": 34702 },
18  "crc16": 57117
19 }

```

Why it matters. Send-All may stream raw blocked indices first. The later certificate—not the raw indices—is the replayable proof.

Greedy-Probe — CUT success (seed 145) Same as CertTalk above on this clean seed (three messages: CUT_PROPOSE, CUT_CERT, ACK). Probes appear only when needed.

Responder-MinCut — CUT success (seed 145)

```

1 {
2   "t": "SCHEMA", "s": "v1", "q": 0,
3   "p": { "cut_enc": "DELTA16_v1", "path_enc": "RLE4_v1", "s": [0,0], "size": [10,10],
4         "t": [9,9] },
5   "c": 64959
6 }
7 {
8   "t": "CUT_CERT", "s": "v1", "q": 0,
9   "p": { "cs": "BwAHAAEAAQD9Af8B", "k": 5, "digest16": 4126, "w": "Hw==", "sb": ["B"]
10         },
11   "c": 32263
12 }
13 {
14   "t": "NACK", "s": "v1", "q": 1,
15   "p": { "n": "CUT_PROPOSE", "reason": "BLOCKED", "x": [[7,7],[9,7],[6,8]] },
16   "c": 593
17 }
18 {
19   "t": "PROBE", "s": "v1", "q": 1,
20   "p": { "cs": [[7,7],[9,7],[6,8]], "wht": "CELLS" },
21   "c": 55063
22 }
23 {
24   "t": "HELLO", "s": "v1", "q": 2, "p": {}, "c": 26486
25 }
26 {
27   "t": "DONE", "s": "v1", "q": 2, "p": {}, "c": 22322
28 }

```

Why it matters. Shows the longer path to agreement when the first cut is insufficient; one small probe reconciles the disagreement.

6 Results in One Table

Medians across 1,000 cached instances (schema-free run).

System	Bytes	Rounds
Send-All	364	3
CertTalk	468	5
Greedy-Probe	468	5
Responder-MinCut	506	6

Takeaway: CertTalk spends about one extra hundred bytes versus Send-All, but those bytes *are the proof* (path or cut), which simplifies auditing.

7 Limits and What to Expect

- Two agents; 10×10 grids; reliable in-order channel. Larger grids may require chunked certificates.
- No privacy model: witness bits reveal who attested each cut cell. Signatures are optional if authenticity is required.
- The checker never reads the transcript; it only replays the final artifact with U .